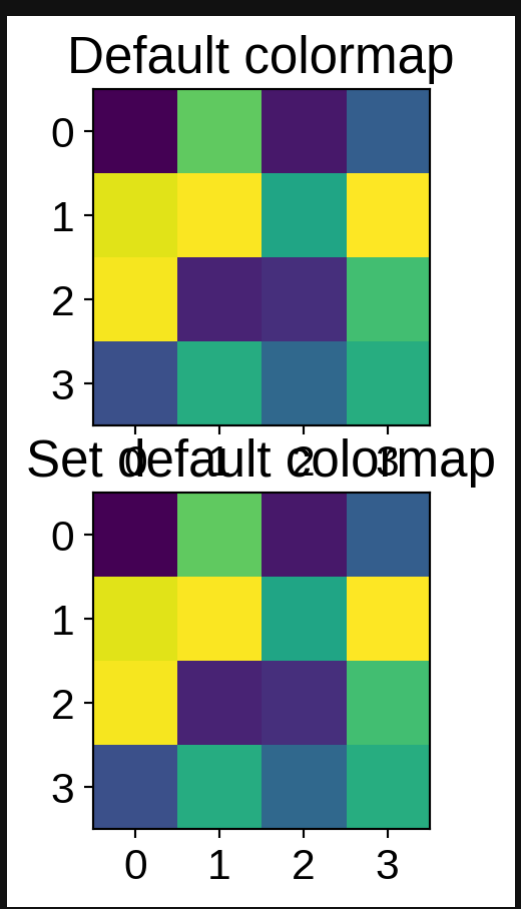
CPSC 330 Lect
DBSCAN
Hierarchic
Clusterin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iClicker Exercise 15.1

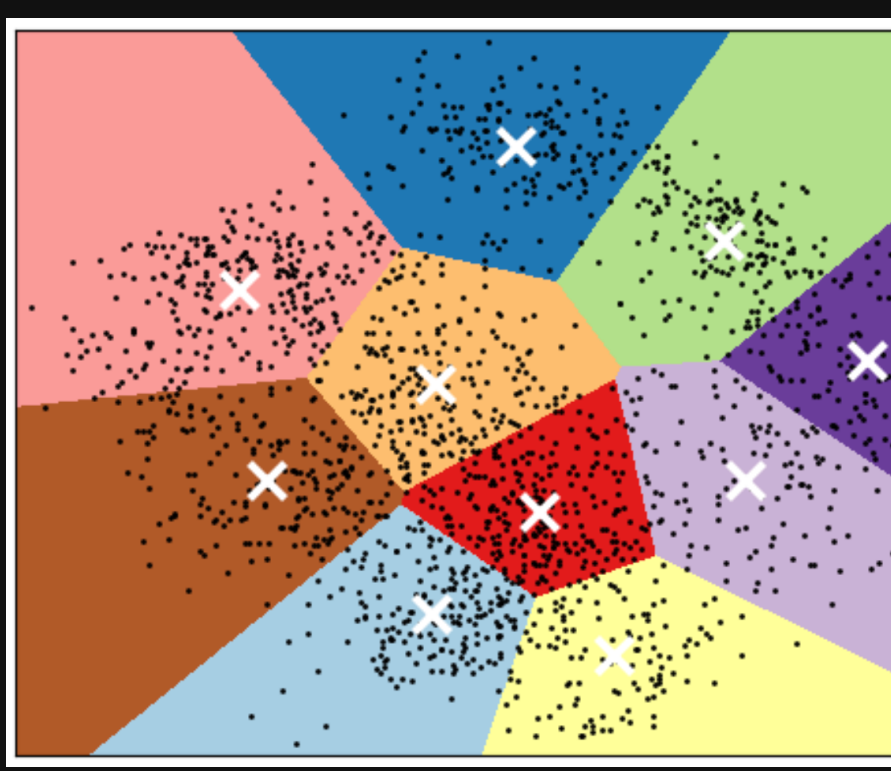
Select all of the following statement TRUE.

- A. With *n* examples, *K* clusters, and *d* Means learns *K* cluster centers, ea
- B. The meaning of *k* in k-nearest neig Means clustering is very similar.
- C. Scaling of input features is crucial i
- D. In clustering, it's almost always a g equal-sized clusters.

Limitations of
means

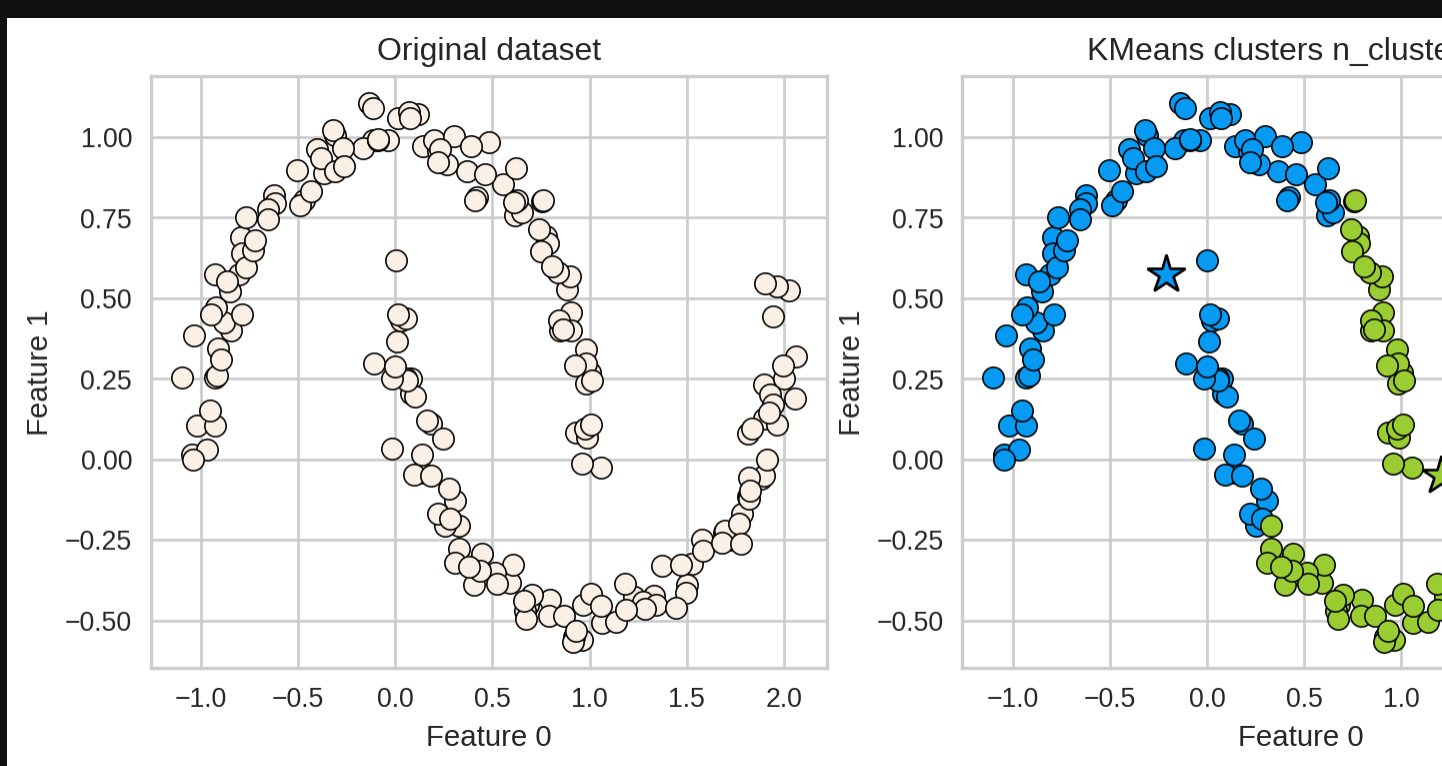
Shape of clusters

- Good for spherical clusters of more o



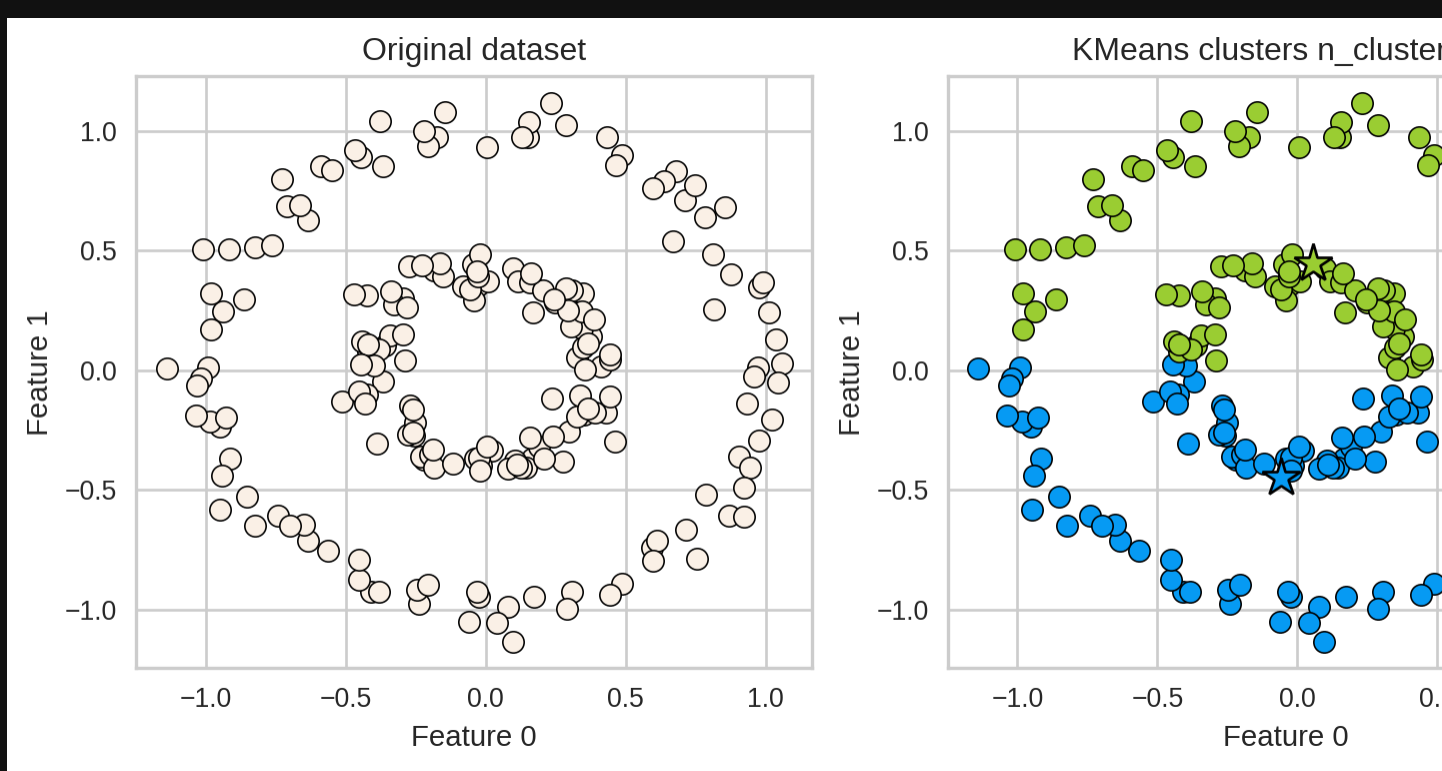
K-Means: failure case 1

- K-Means performs poorly if the clust
  complex shapes (e.g., two moons da



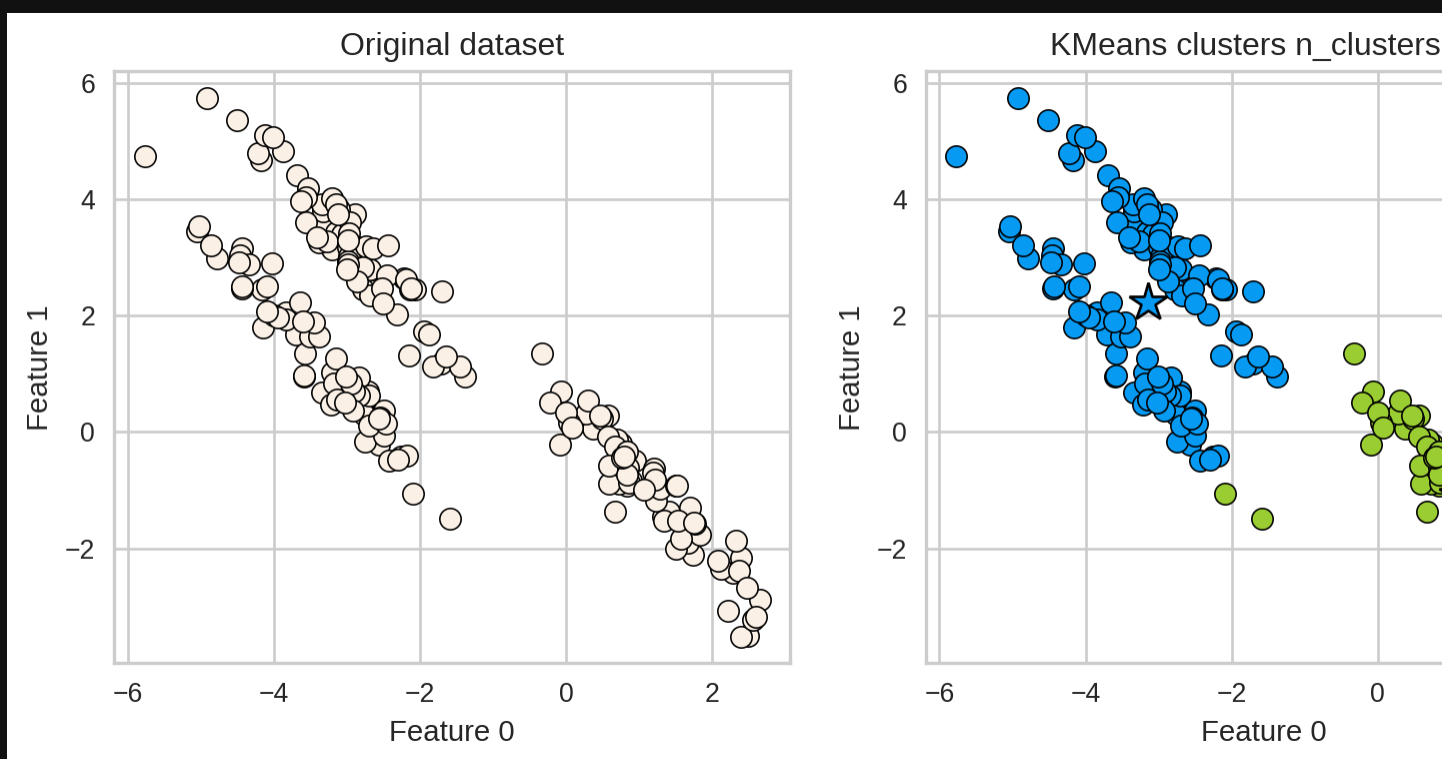
K-Means: failure case 2

- Again, K-Means is unable to capture shapes.



K-Means: failure case 3

- It assumes that all directions are equ
  for each cluster and fails to identify n
  clusters.

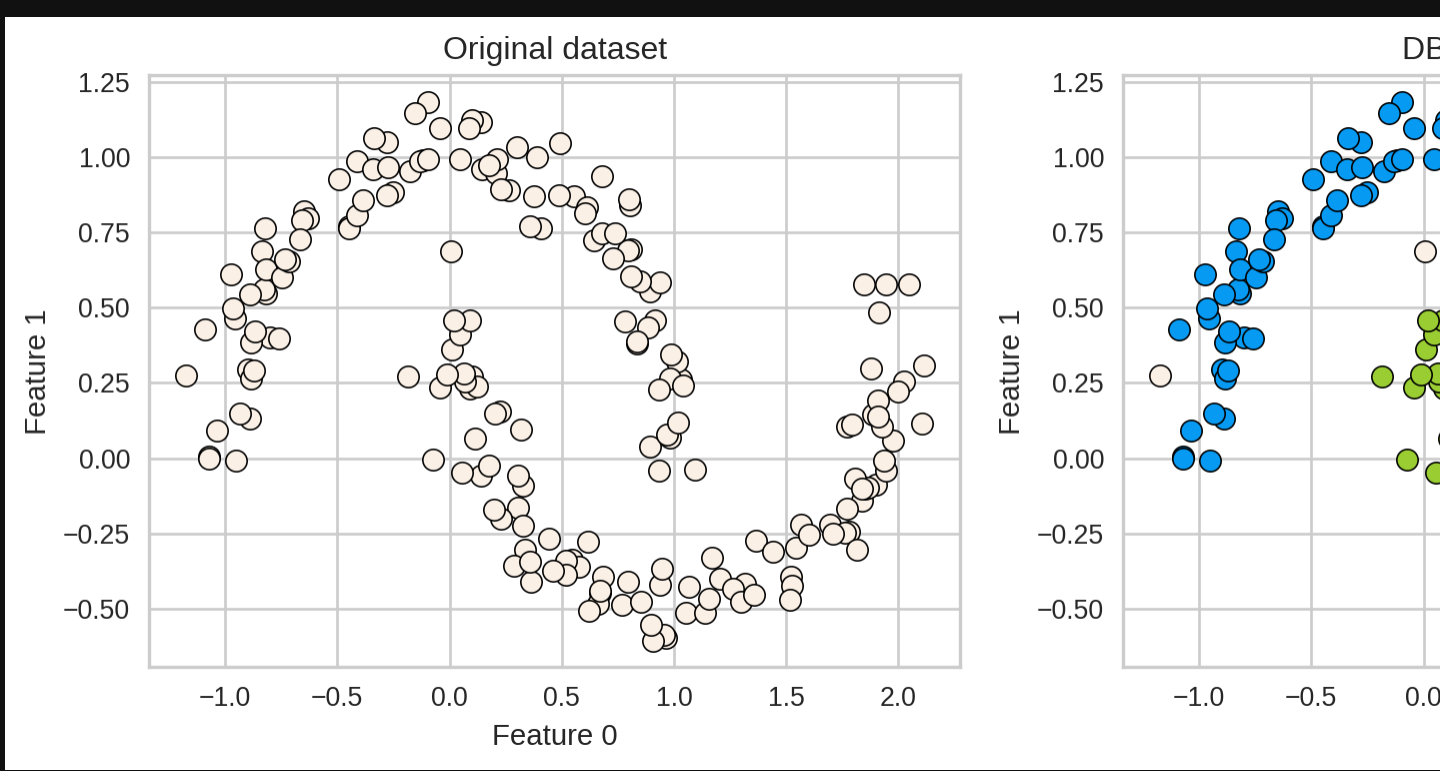


Can we do bet

DBSCAN

- **D**ensity-**B**ased **S**patial **C**lustering of **A**pplications with
- A density-based clustering algorithm

```
1  X, y = make_moons(n_samples=200, noise=0.08, random_state=42)
2  dbscan = DBSCAN(eps=0.2)
3  dbscan.fit(X)
4  plot_original_clustered(X, dbscan, dbscan.labels_)
```

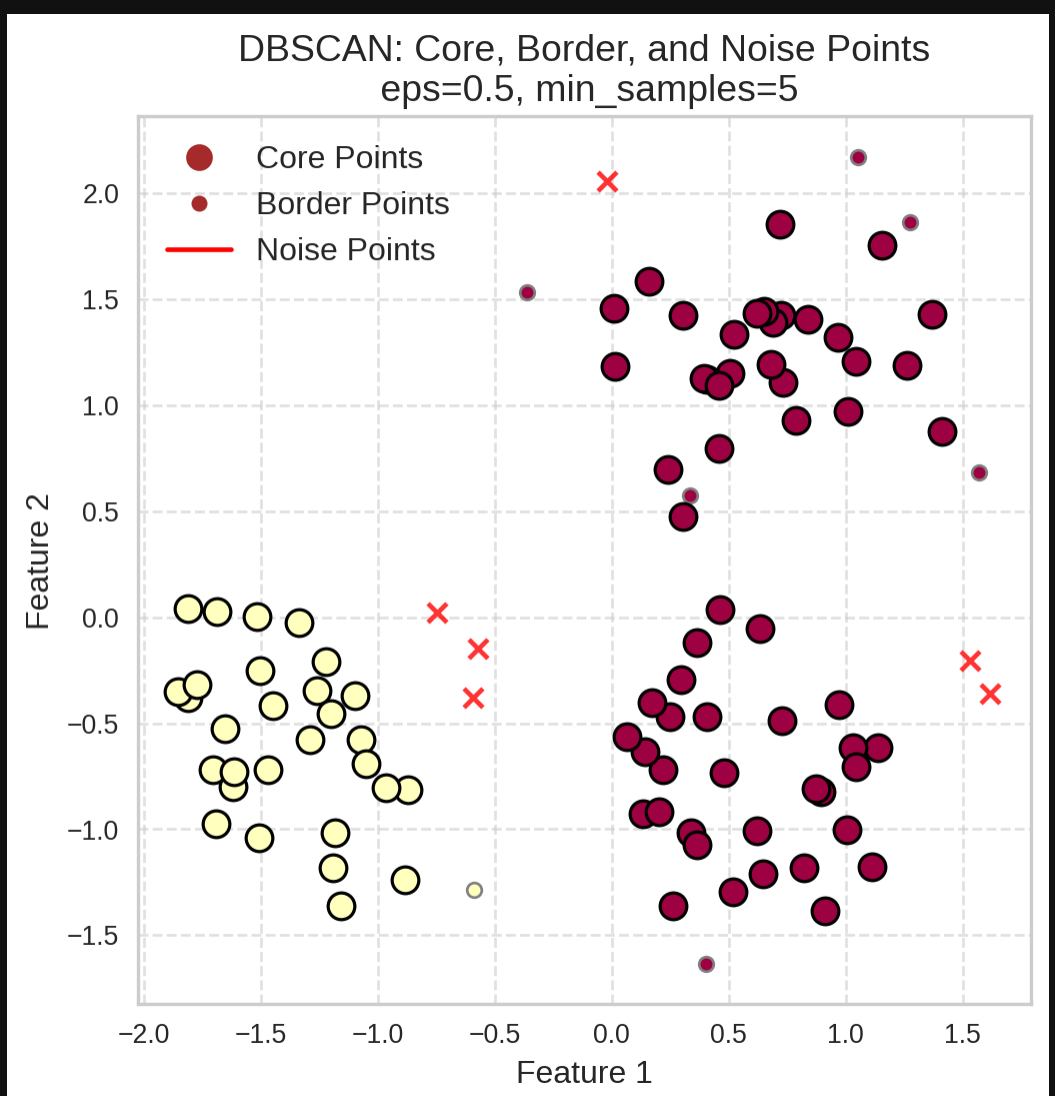


Two main hyperparam

In order to identify dense regions, we n
hyperparameters:

- `eps`: determines what it means for po
- `min_samples`: determines the numbe
  neighbouring points we require to
  for a point to be part of a cluster

DBSCAN analogy



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DBSCAN algorithm

- Pick a point *p*
- Check wheth not.
- If *p* is a core (label).
- Spread the c neighbours.
- Check if any received the yes, spread t neighbors as
- Once there a left to spread unlabeled po

Clustering Algorithm V

Here is a really nice web-app to visualiz

clustering-visualizer.web.app/dbscan

🧩Human DBSCAN act

Goal

Experience how ***ε*** **(eps)** and **min_samp**
counts as a cluster.

Setup

- Each student = one **data point**
- Distance (eps) = **arm's length (~1 m)**
- Each gets a **sticky note** to mark clus
  marker (colour)
- Start from a random student: check

Two parallel runs

Side of Class	ϵ (eps)	min_samples
Left side	~1 m (arm's length)	5
Right side	~1 m (arm's length)	15

Side of Class	ϵ (eps)	min_samples

How to play

1. Pick a **starting student**.
2. Count how many of your neighbours
   distance.
3. If you have ≥ **min_samples**, you are
   start spreading your colour.
4. Neighbours who receive a colour but
   border points.
5. Students who never receive a colour
   points.

After the activity

- Which side formed **more clusters**?
- What happened when **min_samples**
- Why doesn't DBSCAN need to know t clusters *k*?

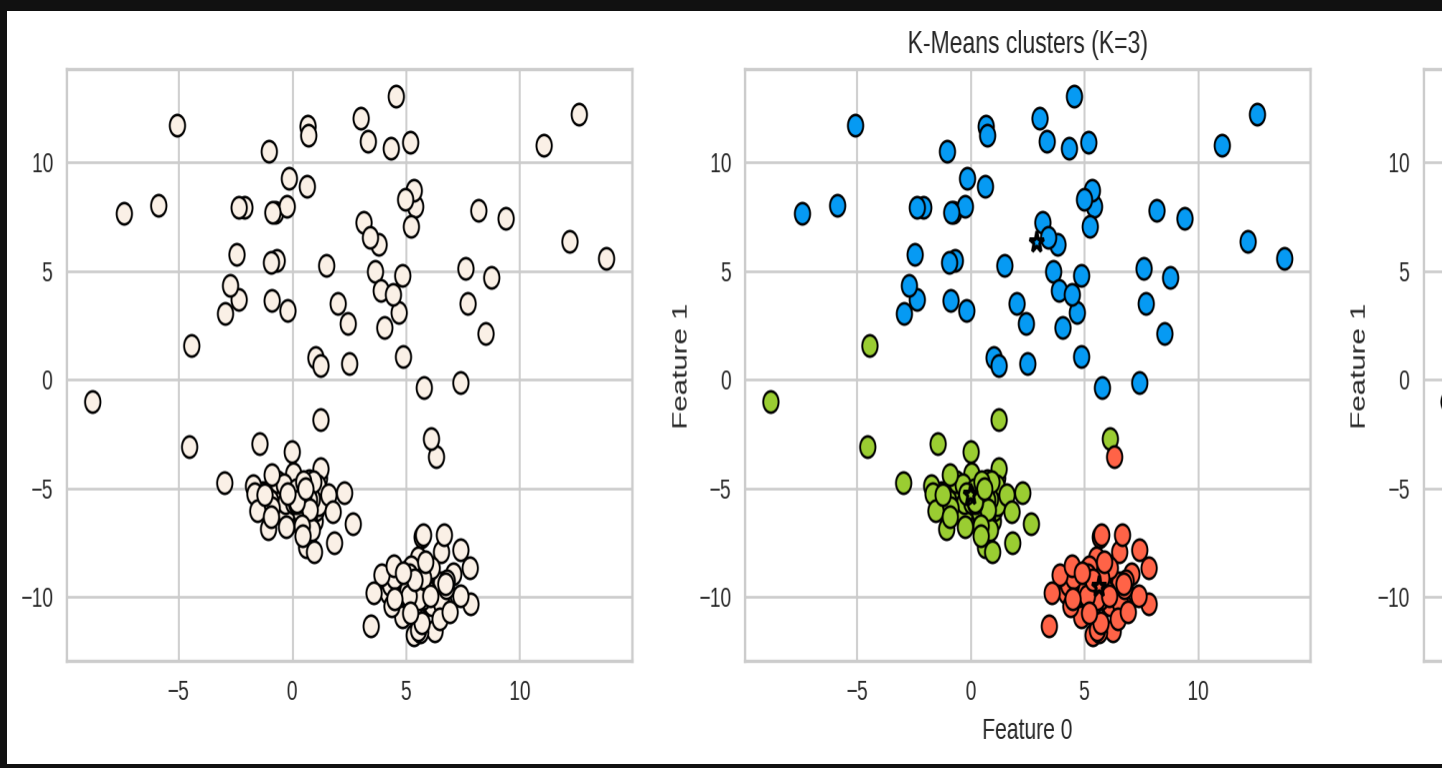
How to tune `eps` and `mi`

- Can you use `GridSearchCV` and `Rando`

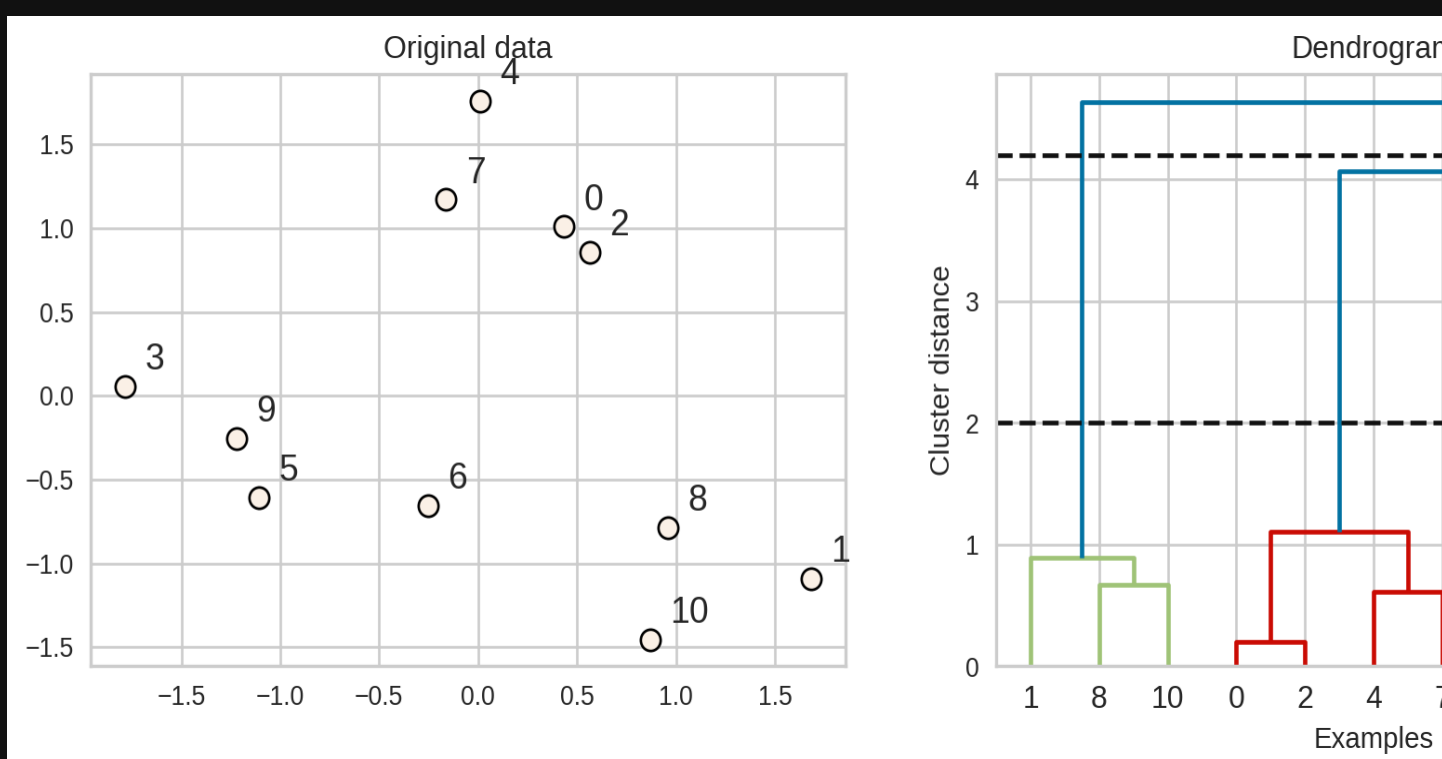
DBSCAN: failure cases

- Let's consider this dataset with three clusters of varyi
- K-Means performs better compared to DBSCAN. But knowing the value of *K* in advance.

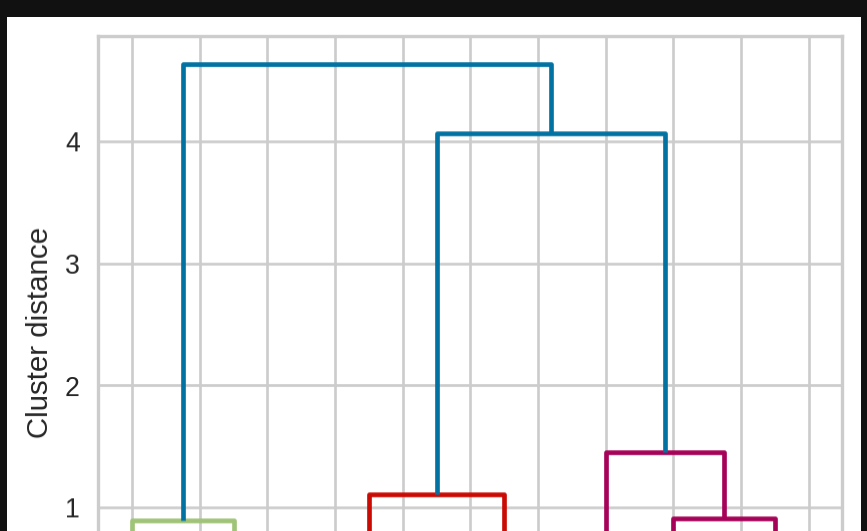
```
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15]
```



Hierarchical clustering



Dendrogram



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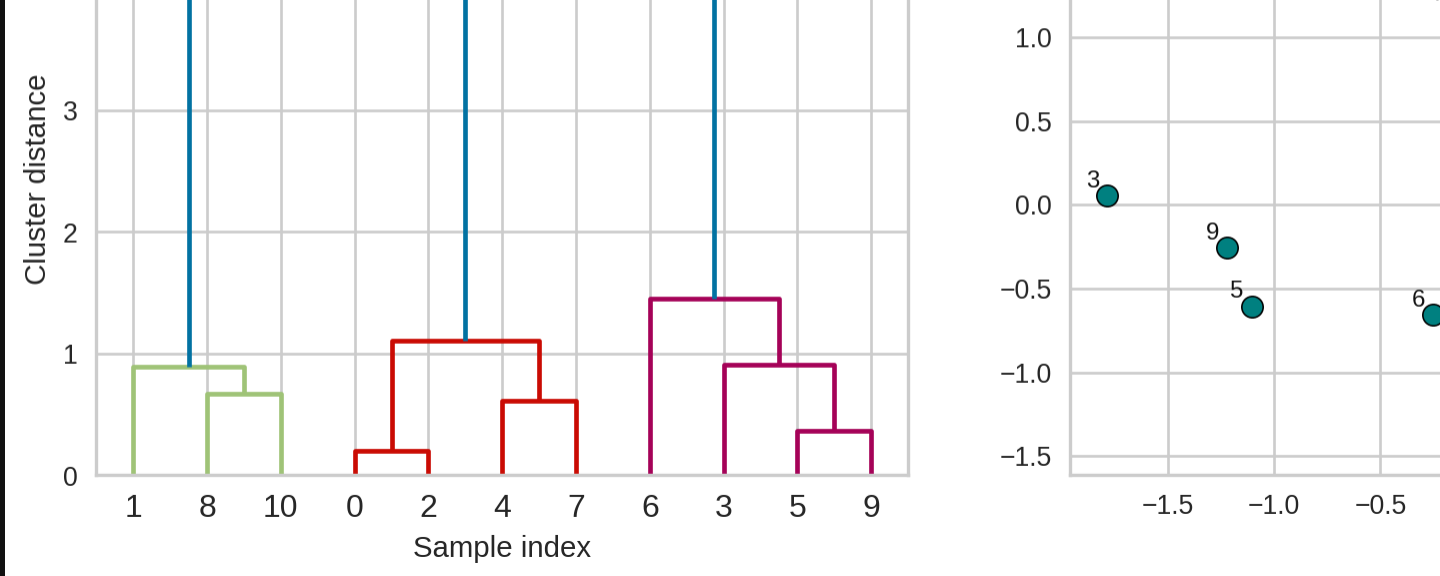
Flat clusters

- This is good but how can we get clus
  dendrogram?
- We can bring the clustering to a “flat”
  `fcluster`

Flat clusters

```
1 from scipy.cluster.hierarchy import fcluster
2 # flattening the dendrogram based on maximum numbe
3 hier_labels1 = fcluster(linkage_array, 3, criteric
4 plot_dendrogram_clusters(X, linkage_array, hier_la
```



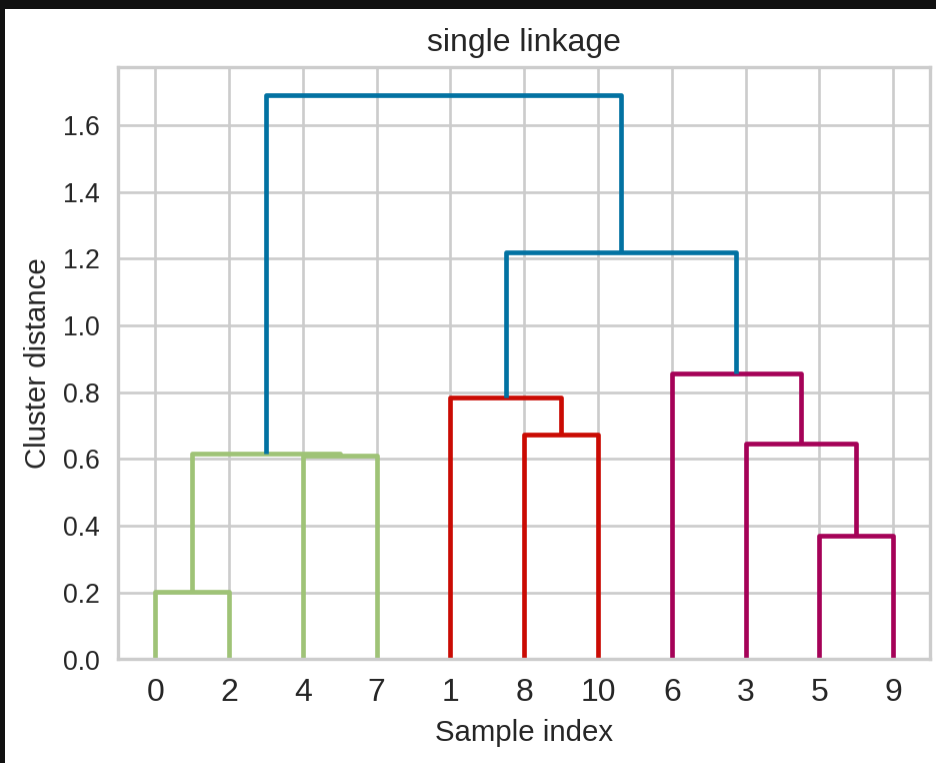
Linkage criteria

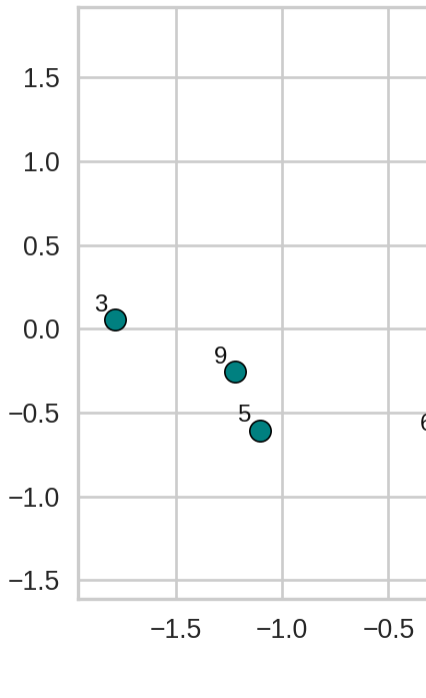
- When we create a dendrogram, we need to calculate
  clusters. How do we measure distances between clus
- The **linkage criteria** determines how to find similarit
- Some example linkage criteria are:
  - Single linkage → smallest minimal distance, leads
  - Complete linkage → smallest maximum distance,
  - Average linkage → smallest average distance betw
    the clusters
  - Ward linkage → smallest increase in within-cluster
    equally sized clusters

Example: Single linkag

Suppose you want to go from 3 clusters to 2 clusters. W
merge?

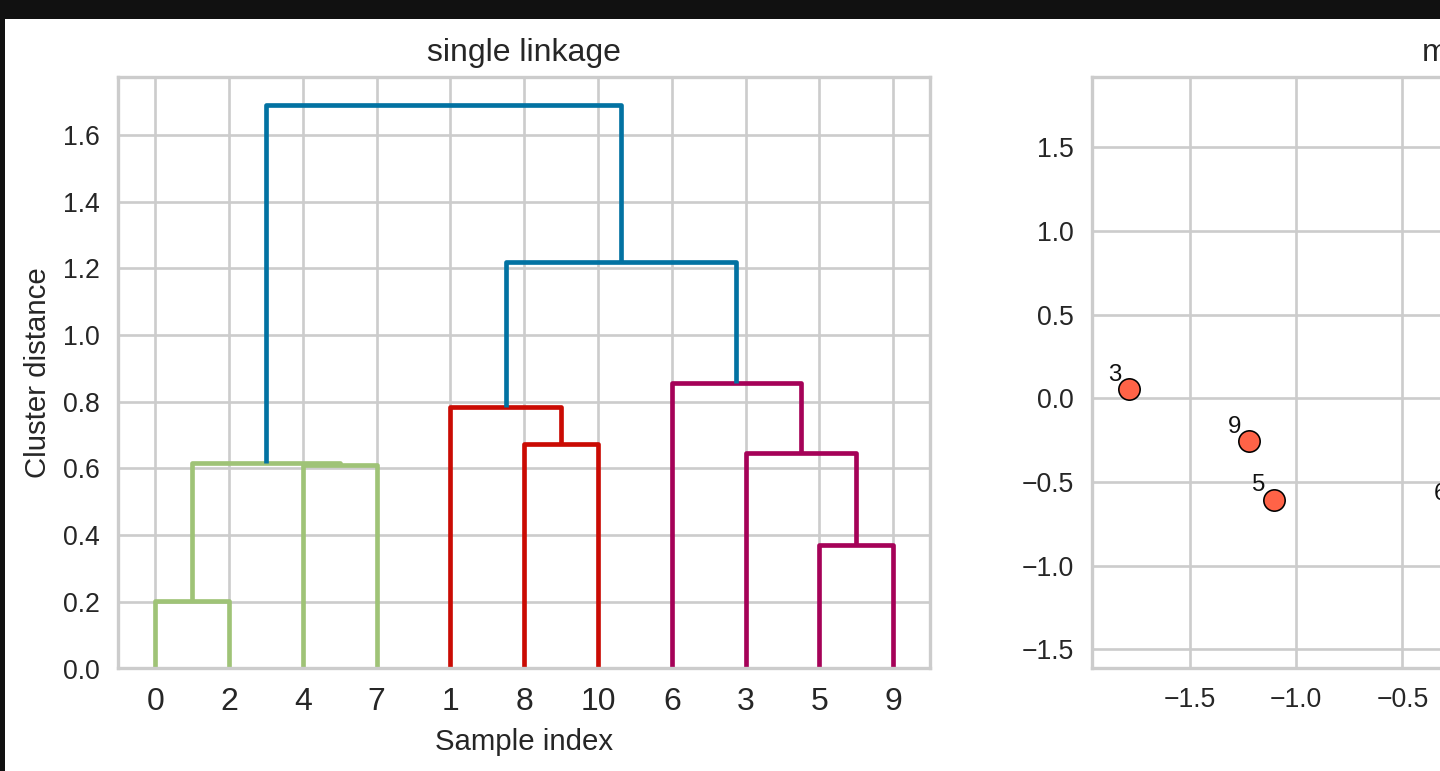
```
1  X_orig, y = make_blobs(random_state=0, n_samples=11)
2  X = StandardScaler().fit_transform(X_orig)
3  linkage_array = single(X)
4  hier_labels = fcluster(linkage_array, 3, criterion="maxclust")
5  plot_dendrogram_clusters(X, linkage_array, hier_labels, title="maxclust 3
```





Example: Single linkag

```
1 hier_labels = fcluster(linkage_array, 2, criterion="maxclust")
2 plot_dendrogram_clusters(X, linkage_array, hier_labels, title="maxclust 2
```



iClicker Exercise 15.2

Select all of the following statements which are Tru

- A. In hierarchical clustering we do not have to worry
- B. Hierarchical clustering can only be applied to smal
  dendrograms are hard to visualize for large datase
- C. In all the clustering methods we have seen (K-Mea
  clustering), there is a way to decide the number of
- D. To get robust clustering we can naively ensemble
  the most popular label) produced by different clus
- E. If you have a high Silhouette score and very clean
  means that the algorithm has captured the seman
  of our interest.

Activity

Discuss the following

Clustering Method	KMeans	DBSC
Approach		
Hyperparameters		
Shape of clusters		
Handling noise		
Distance metric		

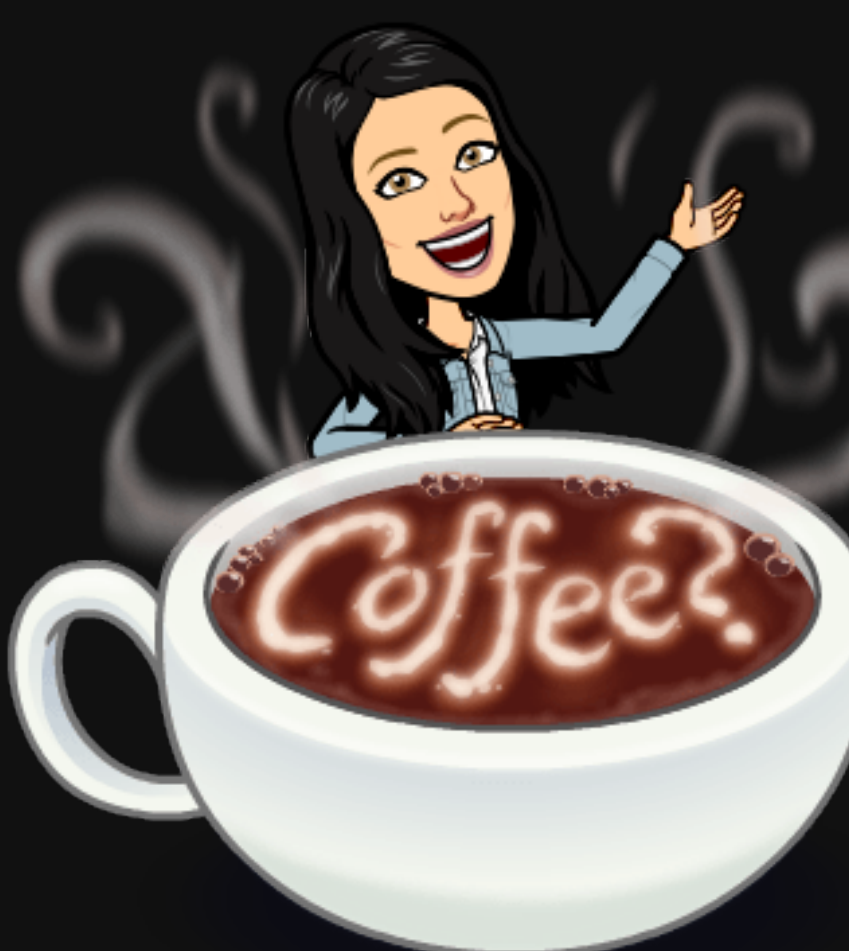
Discussion question

Which clustering method would you us
scenarios below? Why? How would you
data in each case?

- Scenario 1: Customer segmentation
- Scenario 2: An environmental study a
clusters of a rare plant species
- Scenario 3: Clustering furniture items
management and customer recomm

Break

Let’s take a break!



Group Work: Class Dem
Coding

Super cool Demo!

For this demo, each student should clic
create a new repo in their accounts, th
repo locally to follow along with the de

All credit to Dr. Varada Kolhatkar for pu
together!